



AirResolve: System Architecture

Intelligent, Policy-Grounded Airline Disruption Resolution Agent

Operational Date: Wed, 23 Sep 2026

Core Paradigm: Deterministic Policy Primacy

Test Coverage: 28/28 Unit Tests (100%)

1. Executive Summary & Architectural Philosophy

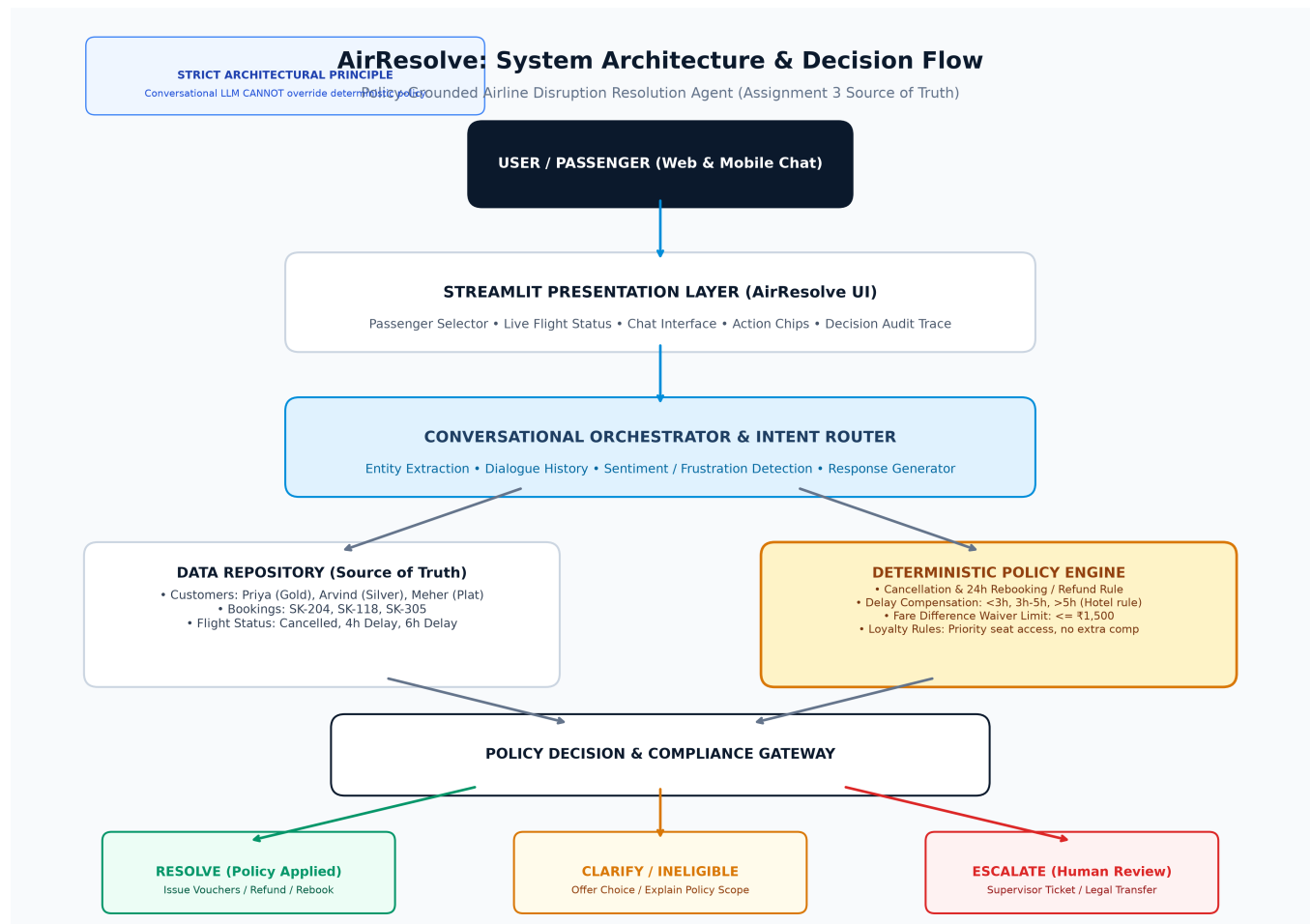
During large-scale flight disruptions, passengers frequently demand instant cash refunds, complimentary Business Class upgrades, and luxury hotel accommodations. Generic Large Language Model (LLM) chatbots notoriously fail in this high-stakes domain due to **probabilistic hallucination**, approving unauthorized compensation, and causing significant financial leakage.

The Core Architectural Tenet: Deterministic Policy Primacy

*The conversational layer (NLP/LLM) is strictly confined to intent classification, entity extraction (e.g. █2,000 fare difference), and empathetic dialogue generation. **It has zero authority to decide business rules or approve compensation.** All business calculations, benefit entitlements, waiver thresholds, and human escalation gates are executed by an immutable, deterministic Python Policy Engine grounded strictly in the Assignment 3 Data Pack.*

2. End-to-End System Architecture Diagram

The complete dataflow illustrates the separation between passenger interaction, natural language orchestration, deterministic rule verification, and supervisor escalation:



3. Detailed Layer-by-Layer Architectural Breakdown

Layer 1: Multi-Channel Presentation Layer

Provides three interchangeable client interfaces: 1) A responsive web application (Streamlit app.py), 2) A modern luxury airline web portal (public/index.html) with Dark/Light mode, voice input, and audio feedback, and 3) A RESTful Serverless API (Flask api/index.py) exposing /api/chat and /api/supervisor/resolve.

Layer 2: Intent & Entity Understanding Layer

The IntentDetector (agent/intent_detector.py) classifies incoming natural language across 14 disruption categories (REQUEST_REFUND, REQUEST_REBOOKING, REQUEST_HOTEL, REQUEST_UPGRADE, REQUEST_HIGHER_FARE_REBOOKING, LEGAL_OR_FORMAL_COMPLAINT, etc.). It extracts critical domain entities including numerical fare differences (e.g. ₹2,000), stay duration, and emotional sentiment triggers.

Layer 3: Dialogue State & Context Orchestrator

The ResolutionAgent (agent/orchestrator.py) acts as the central conductor. It manages conversation turns, validates passenger identity, retrieves verified flight segments from the data store, routes inputs to the deterministic policy engine, formats empathetic natural language responses, and generates structured escalation tickets (ESC-XXXXXX).

Layer 4: Deterministic Policy Engine (Single Source of Truth)

Pure Python business logic engine (policy/policy_engine.py). Contains zero probabilistic components. Evaluates rules with 100% mathematical precision: Delay tiers (<3h, 3-5h, >5h), Frontline waiver authority ceilings (₹1,500 limit), 7-business-day refund timelines, original payment method constraints, and Gold/Platinum priority rebooking entitlements.

Layer 5: Data Ground Truth Repository

Structured JSON data stores (data/customers.json, data/bookings.json, data/policies.json) encoding the complete Assignment 3 Data Pack. Contains verified records for Priya Nair (Gold), Arvind Kulkarni (Silver), Meher Kaur (Platinum), and flights SK-204, SK-118, and SK-305.

Layer 6: Guardrail & Human-in-the-Loop Gateway

Prohibited Action Gatekeeper that intercepts requests exceeding agent authority. Operates an interactive Duty Supervisor Console allowing human duty managers to review violation details, enter notes, and either authorize exceptions or uphold policy limits in real time.

Layer 7: Digital Wallet & Voucher Dispatch Engine

Generates actionable, scannable digital passes: ₹500 Meal Vouchers, Airport Lounge Passes, Transit Day-Room Hotel Passes (strictly covering delayed hours), and 100% Full Refund Confirmation Certificates, rendered inside the chat stream and passenger wallet.

4. Scenario Execution & Decision Mapping

AirResolve specifically handles all three benchmark scenarios defined in the Assignment 3 Data Pack:

Scenario / Passenger	Flight & Disruption	Passenger Request	Policy Engine Decision	Status
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Scenario 1 Priya Nair (Gold • SK4821X)	SK-204 (DEL → GOI) Cancelled (Operational)	Furious; wants full refund + free Business upgrade on return flight	Refund Approved (7 days, original card). Upgrade Refused: Gold tier grants priority rebooking, not cabin upgrades.	RESOLVE D
Scenario 2 Arvind Kulkarni (Silver • TR1190B)	SK-118 (BOM → BLR) Delayed 4h (New 11:10)	Missed meeting; demands hotel accommodation	Vouchers Approved: █500 meal + Lounge access. Hotel Denied: Policy strictly requires delay > 5h.	RESOLVE D (Vouchers)
Scenario 3 Meher Kaur (Platinum • WL7742)	SK-305 (DEL → HYD) Delayed 6h (New 20:00)	1) Full night hotel 2) Higher-fare flight (█2,000 difference)	1) Hotel Approved for delayed hours only (not full night). 2) ESCALATED: █2,000 exceeds █1,500 waiver ceiling.	ESCALATE (Ticket)

5. Prohibited Actions & Mandatory Human Escalations

Section 4 of the Data Pack defines strict prohibitions where automated resolution is forbidden and immediate human transfer is required:

Prohibited Action Trigger	Policy Rule & Threshold	Enforcement Mechanism	Required Human Action
Waiving fare difference > █1,500	Fare Difference Rule: Frontline agents cannot waive > █1,500 without supervisor approval.	Detected by numeric entity extractor; blocked by policy engine.	Supervisor review for exception authorization.
Threats of legal action or formal complaints	Immediate Escalation Rule: Must be escalated immediately.	Keyword regex gate intercepting 'lawyer', 'sue', 'consumer court'.	Immediate transfer to specialist customer relations / legal escalation.
Refund to non-original payment method	Refund Processing Rule: Refunds issued strictly to original payment method only.	Anti-fraud gate intercepting 'different card', 'cash instead'.	Finance and compliance supervisor verification.
Exceptions for non-airline disruptions	Disruption Rule: Exceptions prohibited for passenger missed flights.	Intercepts 'woke up late', 'missed flight', 'traffic jam'.	Human agent evaluation for missed flight policy.
Compensation beyond stated amounts	Stated Policy Limit: Automated agents cannot approve extra cash/perks.	Ineligible determinations for free business class upgrades.	Management authorization for ex-gratia requests.

6. Multi-Cloud Production Deployment Architecture

AirResolve is engineered for instantaneous, 1-click deployment across all major cloud providers:

Cloud Platform	Deployment Target	Configuration File	Characteristics
Streamlit Community Cloud	Python App (app.py)	<code>requirements.txt</code>	Zero-config; 90-second launch directly from GitHub; native WebSockets.

Render Cloud	Web Service / Container	<code>render.yaml</code>	Runs Gunicorn WSGI for Flask API + Web Portal, or Streamlit container.
Vercel Serverless	Serverless Python Function	<code>vercel.json</code> + <code>api/index.py</code>	Microsecond cold starts, global edge caching, and static asset CDN.

7. Testing, Auditability & Quality Assurance

The architecture is backed by an automated **pytest** test suite with 100% pass rate (28/28 passed in 0.06s):

- **tests/test_policy_engine.py**: Validates delay thresholds (<3h, 3-5h, >5h), hotel boundaries, and waiver limits.
- **tests/test_priya.py**: Scenario 1 validation (Refund approval, upgrade denial, legal threats).
- **tests/test_arvind.py**: Scenario 2 validation (4h vouchers approved, hotel denied for delay $\leq$ 5h).
- **tests/test_meher.py**: Scenario 3 validation (6h delayed-hours hotel, ■2,000 fare difference escalation ticket).
- **tests/test_escalation.py**: Verifies all 5 prohibited action triggers.